Fully Symmetry-Conditioned Rigid-Body Flow Matching for Molecular-Crystal Structure Prediction

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Abstract

Rigid-body flow matching reduces molecular-crystal structure prediction to a lattice, fractional centroids, and per-molecule orientations on SO(3), but current rigid-body molecular-crystal generators leave crystallographic symmetry unconditioned, and symmetry-conditioned generation has been demonstrated only for inorganic atomic sites. We show that the per-molecule orientation target factorizes as $R_m = \mathrm{rot}(g_m)\, R_{\mathrm{asym}}$ into a space-group-determined relative rotation, which a flow learns, and a gauge-free asymmetric-unit pose, which it cannot regress from packing. We turn this into a deployable method: a leak-free coset label—the generating space-group operation, recoverable from a symmetry template at sampling time—conditions the orientation flow. On Cambridge Structural Database crystals, deployable coset conditioning lifts the held-out non-reference orientation loss by 41.1% (versus 27.5% for a paired control), two-thirds of the way to a leaky-codebook oracle ($\sim 48\%$); an SO(3)-averaged objective closes the gap (47.7%). Supplying the coset at generation collapses the end-to-end orientation error sevenfold ($93.5^\circ \to 13.4^\circ$), and the advantage widens with data. Extending symmetry conditioning to the lattice—a crystal-family mask on an O(3)-invariant log-metric parametrization—gives, to our knowledge, the first fully symmetry-conditioned molecular-crystal flow. Paired with an unsupervised symmetry-preserving packing finisher, a fully-ablated stack of levers lifts held-out exact match from 0% to 6.9% (strict best-of-10, StructureMatcher stol 1.0), at statistical parity with the symmetry-free MolCrystalFlow ($\sim$ 8%), reaching 10.7% with orientation test-time augmentation. We report the contribution of each lever.

Introduction

Crystal structure prediction (CSP) seeks the stable periodic packing of a chemical system, and generative models have become a fast alternative to large ensembles of density-functional relaxations, progressing from crystal diffusion autoencoders[1] to joint equivariant diffusion[2] and Riemannian flow matching,[3–5] which traverses a straight conditional path solvable in tens of steps. Rigid-body factorization is especially attractive for molecular crystals: freezing each molecule's conformer collapses dozens of atomic coordinates to a single pose—a lattice L, a fractional centroid $c_m \in \mathbb{T}^3$, and an orientation $R_m \in \mathrm{SO}(3)$ per molecule—and supports the short sampling trajectories characteristic of flow matching.

The gap. This factorization introduces a per-molecule orientation variable on SO(3), and here the current generation of models underuses the available symmetry. MolCrystalFlow,[6] a 2026 rigid-body molecular-crystal flow with the identical lattice $\times\, \mathbb{T}^3 \times \mathrm{SO}(3)$ factorization, jointly learns pose but imposes no space-group constraint, and MOFFlow[7] flows rigid SE(3) poses relying on framework connectivity rather than a symmetry prior. Symmetry- and Wyckoff-conditioned generation, meanwhile, is proven only for *inorganic atomic sites*: DiffCSP++,[8] WyckoffDiff,[9] SymmCD,[10] and the 2026 NextCrystal[11] all condition atom generation on a space-group/Wyckoff template, but none conditions rigid-body molecular orientation. We close that gap and make the conditioning *deployable*.

Contributions.

- A leak-free, template-available coset label (the generating operation $h = g_m g_0^{-1}$, recovered from centroids alone) that conditions the orientation flow without seeing the target.
- Deployable conditioning recovers two-thirds of the oracle-codebook orientation benefit (+41.1% vs a +27.5% paired control); an SO(3)-averaged objective closes the rest (+47.7%).
- End-to-end, supplying the coset collapses the orientation error 7× ($93.5^\circ \rightarrow 13.4^\circ$), and the coset advantage *widens* with data.
- A template-free path: a packing-only predictor whose top-k covers the true coset for 86% of molecules at k=5, made usable by top-k marginalization.
- Extending symmetry conditioning to the lattice (a crystal-family mask on a log-metric parametrization) gives the first *fully symmetry-conditioned* molecular-crystal flow; composed with an unsupervised packing finisher and a fully-ablated lever stack, it lifts exact match from 0% to 6.9% (strict best-of-10, stol 1.0)—parity with MolCrystalFlow—and 10.7% with orientation test-time augmentation.

Methods

Rigid-body flow

SymMC-Flow represents a molecular crystal by a lattice $L \in \mathbb{R}^{3\times3}$ and, per molecule m, a fractional centroid $c_m \in \mathbb{T}^3$ and an orientation $R_m \in \mathrm{SO}(3)$ acting on a shared body-frame conformer, so that atom a of molecule m sits at $\mathrm{cart}_{m,a} = c_m L + R_m\, \mathrm{local}_{m,a}$. Each crystal is parsed into rigid blocks by a covalent-bond graph with periodic unwrapping; species are keyed by a Weisfeiler–Leman hash and share one reference conformer rotated into a molecule-intrinsic frame, so each copy's pose is the Kabsch rotation to that frame (Supporting Information). A conditional flow-matching objective[4] is trained jointly over three manifolds—a Euclidean path for a volume-normalized log-metric lattice, a torus geodesic for centroids, and an SO(3) geodesic for orientations—with the velocity field conditioned on the space group; an E(n)-equivariant encoder[12] embeds the conformer and pair-bias attention mixes molecules within the cell. Sampling integrates the three coupled fields with an RK4 solver in 50 steps.

The orientation decomposition

A molecule m generated from the asymmetric unit by a space-group operation with rotation part $\mathrm{rot}(g_m)$ has absolute pose

$$R_m = \mathrm{rot}(g_m)\, R_{\mathrm{asym}}, \tag{1}$$

where R_{asym} is the orientation of the asymmetric unit in the cell. The first factor is fixed by the space group and shared across crystals; the second is a free, per-crystal choice. Because the body-frame gauge is arbitrary, R_{asym} carries no consistent signal tied to composition or packing: trained directly on R_m, an absolute-orientation head never leaves its predict-zero floor ($\mathbb{E}\|u_R\|^2 \approx 5.24$) while the lattice and centroid heads converge normally. The empirical distribution is consistent with this—absolute reference-copy orientations track the Haar-uniform density on SO(3) (mean geodesic angle 122.6° against the Haar expectation 126.5°; Kolmogorov–Smirnov $D = 0.04$), and a supervised probe trained to predict R_{asym} from lattice, centroid arrangement, conformer, and space group does not beat the best feature-free constant (Supporting Information). We therefore state the claim operationally and narrowly: R_{asym} is not predictable from the conditioning available here, not that orientation is unlearnable in principle—a richer descriptor such as hydrogen-bond topology or dipole alignment could in principle expose residual signal.

To isolate the learnable factor we re-gauge each crystal so the first copy of every species is

Table 1: Held-out non-reference orientation loss (untrained $\rightarrow$ trained) across orientation target and conditioning. The absolute pose floors; the relative (symmetry-determined) orientation is learnable and robust to conditioning noise.

target	noised conditioning	clean packing
absolute R_m	$5.37 \rightarrow 5.18$ (+3.3%)	$5.35 \rightarrow 5.18$ (+3.3%)
relative R'_m	$5.37 \rightarrow 3.91$ (+27.1%)	$5.37 \rightarrow 3.94$ (+26.7%)

a reference ($R = I$) and the remaining copies carry only the relative rotation $R'_m = R_m R_0^{-1}$, which cancels R_{asym} and leaves the reconstructed crystal unchanged. Splitting the orientation loss into reference copies (target I) and non-reference copies (the symmetry-determined targets) and reporting the latter—which cannot be earned by predicting the identity—the held-out non-reference loss drops by $27.5 \pm 2.7\%$ over three seeds (Table 1), identically under noised and clean conditioning, so the signal rides on the space group (conditioned directly) and the coarse centroid arrangement. A stricter species-grouped split that keeps every crystal sharing a species on one side of the partition gives $28.5 \pm 2.9\%$, ruling out species leakage between training and validation. The floor of the absolute target and the descent of the relative target together confirm Eq. (1): the only learnable content in R_m is the part that survives cancellation of R_{asym}. This is a diagnostic; the coset construction below makes it a deployable method.

Deployable symmetry-coset conditioning

The non-reference relative rotations fall into a discrete set of *cosets*: R'_m equals the rotation part of the operation $h = g_m g_0^{-1}$ that maps the reference copy to copy m. We assign each molecule this generating operation from the crystallography, by matching its fractional centroid against $W_h c_0 + t_h$ (minimum image) over the space group's operations $\{(W_h, t_h)\}$ obtained from pymatgen—never from the observed rotation. The resulting label is **leak-free** and available at sampling time from a symmetry template, exactly as DiffCSP++/WyckoffDiff-style generators supply Wyckoff templates for atoms.[8,9] This yields

238 operation-cosets across the corpus (versus 707 for a codebook clustered from the target rotations, whose index is derived in part from the answer and so serves only as an upper bound). The coset index enters the network as a per-molecule embedding and is threaded through the sampler, conditioning both training and generation.

SO(3)-averaged objective

The SO(3) flow-matching residual has high variance because the prior rotation is drawn uniformly. Following the SO(3)-averaged flow objective for conformer generation,[13] we average the orientation residual over K prior draws per molecule before the gradient step, reducing target variance at fixed data.

Template-free coset prediction

For template-free deployment the coset must be inferred. We train a classifier that predicts each molecule's coset from the conformer, centroid, lattice, and space group—*without* orientation, so it cannot leak the target. Because its argmax is noisy, we *marginalize the top-k*: condition the flow on each of the k most-likely cosets and select the best draw (in deployment by R_{wp}/energy; here, an oracle upper bound).

Full symmetry conditioning: the crystal-family lattice mask

The coset conditions orientation; the lattice remains free. Molecular crystals are overwhelmingly monoclinic and orthorhombic, so reference cells place $\sim 74\%$ of cell angles at exactly $90°$, yet an unconstrained 9-DOF shape head reproduces this for only $\sim 16\%$. We condition the lattice on the crystal family the way DiffCSP++[8] conditions inorganic sites: represent the cell by the O(3)-invariant log-metric $\log(L^\top L) \in \mathbb{R}^6$, whose first coordinate is the volume and whose remaining coordinates are the deviatoric shape directions, and apply a binary *family mask* that freezes the constrained coordinates to their canonical values after each sampler step and in the prior (basis and per-family mask in the Supporting Information). The flow thus learns only the free, family-consistent degrees of freedom. The mask mechanism is DiffCSP++'s; our contribution is transporting it into the molecular rigid-body flow and composing it with the orientation coset, giving a fully symmetry-conditioned flow.

Exact-match finishing: physical press, learned prior, self-conditioning, recycling

Symmetry conditioning is necessary but not sufficient for exact match; a small stack of levers supplies the remaining positioning precision, each ablated independently (§). *(i) Rigid-press finisher:* an unsupervised, symmetry- and rigidity-preserving relaxation of the generated packing—a Lennard-Jones energy over intermolecular atom pairs (van der Waals radii), optimized over the asymmetric-unit pose and the family-masked cell, with symmetry copies regenerated from the space-group operations each step, so it cannot break the imposed symmetry and never sees the target. *(ii) Learned centroid prior:* optimal-transport coupling[14] with a fixed Gaussian centroid prior repairs an under-trained centroid head. *(iii) Self-conditioning:* the model optionally consumes its own terminal estimate of the structure, trained with probability 0.5 and carried across sampler steps. *(iv) Recycling:* AlphaFold-style, the sampling trajectory is re-solved conditioned on the previous pass's output structure. *(v) Orientation test-time augmentation (TTA):* perturbing the asymmetric-unit orientation and pooling the best finished draw. All are gated behind configuration flags, leaving the default generation path unchanged (finisher energy/optimizer and the self-conditioning update in the Supporting Information).

Data and evaluation

The corpus is a seeded CSD sample:[15] 1,127 ordered organic molecular crystals (5,048 rigid blocks, a fixed 964/131 split), with 1,095 crystals containing ≥ 2 copies of a species—the

Table 2: Held-out non-reference orientation-loss drop, mean±s.d. (3 seeds). The deployable coset is leak-free and available from a template; the leaky codebook (index clustered from the target) is an upper bound.

configuration	non-ref. loss drop	deployable?
no-coset control (baseline)	$27.5 \pm 2.7\%$	—
deployable coset (K=1)	$41.1 \pm 3.4\%$	yes
+ SO(3)-averaged	$47.7 \pm 3.9\%$	yes
leaky codebook (oracle upper bound)	$\sim 48\%$	no

multi-copy subset where a relative target exists—and a 2,539-crystal scale corpus for §. The primary training metric is the held-out non-reference orientation-loss drop (percentage reduction from untrained to trained on non-reference copies, which cannot be earned by predicting the identity). We also report orientation-isolated and end-to-end StructureMatcher reconstruction[16] with best-of-k sampling, following match-rate cautions.[17] All learning numbers are means over three seeds.

Results and Discussion

Deployable coset conditioning recovers the oracle benefit

Table 2 gives the main result. A paired no-coset control reproduces +27.5%; the deployable coset reaches +41.1%—two-thirds of the way to the $\sim 48\%$ leaky ceiling—and the SO(3)-averaged objective (K=4) reaches +47.7%, statistically level with the ceiling. The benefit that a target-derived codebook could only *upper-bound* is realized here by a label a sampler actually has.

The coset makes orientation work end-to-end

Sampling all three manifolds from the prior (full de-novo generation), supplying the coset at generation collapses the median orientation error from 93.5° (the same model with the coset withheld) to 13.4°—a 7× reduction—while the centroid error is unchanged (0.346 vs 0.345;

Table 3: Orientation-isolated reconstruction (coset-conditioned, best-of-8; true lattice, centroid, and conformer supplied, only SO(3) sampled): StructureMatcher match rate across the (length, site) tolerance grid on the held-out set. The oracle (true orientation) matches throughout.

(ltol, stol)	$(0.2, 0.2)$	$(0.3, 0.5)$	$(0.5, 0.7)$
coset-conditioned	4.6%	27.5%	37.4%
oracle (true R)	100%	100%	100%

the coset does not touch it) and neither reaches exact match (0%; lattice 0.44 / centroid 0.35 dominate). The coset thus makes the orientation component work end-to-end; joint lattice–centroid generation is the remaining exact-match bottleneck.

Orientation-isolated reconstruction

A drop in flow-matching loss need not correspond to correct geometry. Holding the true lattice, centroid, and conformer fixed and sampling only the coset-conditioned SO(3) field, the flow reconstructs 27.5% of held-out multi-copy packings (best-of-8), rising to 37.4% at looser but oracle-validated tolerances, with a median non-reference orientation error of 41.1°; the oracle (true orientation) matches every structure (Table 3). The symmetry-relative rotations cluster near 180° (the dominant $P2_1/c$, $P\bar{1}$ operations),[15] so a single deterministic integration collapses to the conditional mean and best-of-k is the correct read of a multimodal target.

Template-free deployment via top-k marginalization

The packing-only predictor is 39.5% top-1 (4× the 10.3% majority), but the true coset is in its top-3 for 75.8% of molecules, top-5 for 85.6%, top-10 for 90.9% (Table 4). The signal is present; the argmax merely cannot commit. Marginalizing the top-5 pulls the predicted-coset orientation error from 64.8° (top-1) to 51.2°, toward the 41.1° template ceiling. At the larger corpus (§) the predictor sharpens only to 43.1% top-1 and its predicted cosets still reconstruct below template grade, so closing this residual—a sharper predictor or per-

Table 4: Predictor top-k coverage: true coset in the top k predictions (non-reference molecules, n=438).

top-k	1	3	5	10
coverage	39.5%	75.8%	85.6%	90.9%

molecule combinatorial search—remains the open problem.

Scale

Retraining the full pipeline on a 2,539-crystal CSD corpus (2,464 multi-copy after re-gauge; 2,168 train / 296 validation; 320 distinct operation-cosets), roughly 2.2× the diagnostic set, sharpens every orientation metric while leaving the packing bottleneck untouched (Table 5). The deployable-coset gate does not merely hold: the non-reference loss drop rises to $54.3 \pm 0.8\%$ against a paired no-coset control at $36.8 \pm 2.1\%$, *widening* the coset advantage from +13.6 to +17.5 percentage points—the benefit is a property of the factorization, not a small-data artifact. End-to-end with the coset supplied, the median orientation error falls to 4.3° (from 13.4°), and orientation-isolated reconstruction climbs to 56.8% best-of-8 at 31.7° (from 27.5% at 41.1°). Exact end-to-end match remains 0% with the lattice (0.42) and centroid (0.35) component errors essentially unchanged: scaling data sharpens orientation but does not move the joint lattice–centroid bottleneck.

Comparison to rigid-body baselines

To make the contribution legible to the same reviewers, we re-score held-out generations from the deployable coset model ($N \approx 1.1$k) in MolCrystalFlow's exact units:[6] pymatgen StructureMatcher (ltol=0.3, atol=10°, site tolerance swept 0.5–1.2), best-of-10 match rate, and the relative cell-volume deviation (RMAD) over all ten samples. We pair a de-novo pass (no template, coset off—the MolCrystalFlow/MOFFlow regime) against our symmetry-conditioned pass (coset on) on *identical* prior draws (Table 6).

Table 5: Diagnostic ($N \approx 1.1$k) vs. scale ($N = 2.5$k) corpus: orientation metrics sharpen with data while end-to-end exact match stays lattice/centroid-limited. The gate is the held-out non-reference orientation-loss drop (deployable coset, K=1); orientation-isolated match is best-of-8 with true lattice+centroid supplied.

metric	diagnostic	scale
coset gate: non-ref. loss drop (deployable)	41.1%	**54.3**%
paired no-coset control	27.5%	36.8%
coset advantage	+13.6 pp	**+17.5** pp
end-to-end orientation error (coset supplied)	13.4°	**4.3**°
orient-isolated match (best-of-8)	27.5%	**56.8**%
orient-isolated orientation error	41.1°	**31.7**°
predictor top-1 (majority baseline)	39.5% (10.3%)	43.1% (10.4%)
end-to-end exact match	0%	0%

Two readings are deliberate and honest. First, end-to-end our diagnostic-scale model is *not* competitive with the dedicated flows on either axis: best-of-10 match is at floor (0–0.8%) against MolCrystalFlow's ~6.8% at stol 0.8, and cell-volume RMAD (~31%) lands between MOFFlow (18.8%) and raw Genarris-3 (59%) but well above MolCrystalFlow (3.9%). This is the lattice/centroid bottleneck of Table 5 now measured in the baselines' units: our lattice and centroid heads are diagnostic-grade *before finishing*, and § closes exactly this end-to-end gap—to parity with MolCrystalFlow—with full lattice symmetry conditioning and the finisher. Second, and the point of the paper, is the axis those baselines leave *unconditioned*. The coset barely moves RMAD (32.9% → 30.7%)—it conditions orientation, not packing—but on the same prior draws it collapses the median best-of-10 orientation error from 103° (de-novo) to 14° (conditioned). MolCrystalFlow and MOFFlow are both symmetry-free and have no such lever; MolCrystalFlow is itself only *competitive* with the space-group-conditioned Genarris-3, corroborating that symmetry conditioning—not packing—is the remaining lever. Our contribution supplies exactly that for rigid-body orientation and composes with the strong packing these methods already have. *Caveat:* identical matcher and match@10 as MolCrystalFlow, but our corpus is a smaller CSD-derived molecular set (not Thürlemann/OMC25) at diagnostic scale—a comparable-task alignment, not a strict head-

Table 6: Held-out generations in MolCrystalFlow's units (best-of-10; StructureMatcher ltol= 0.3/atol= 10°; $\mathrm{RMAD_{vol}}$ = relative cell-volume deviation over all 10 samples). Baseline rows are as reported[6,7] (Genarris-3 via the comparison in ref 6). Our end-to-end match and RMAD are lattice/centroid-limited; the coset acts on orientation (last column)—the axis the symmetry-free baselines omit.

method	match@10 stol 0.8	match@10 stol 1.0	$\mathrm{RMAD_{vol}}$ (all 10)	orient. err. (best-of-10)
MolCrystalFlow (sym-free)[6]	∼6.8%	∼8%	3.9%	–
MOFFlow (sym-free)[7]	∼3.4%	–	18.8%	–
Genarris-3 (sym-cond.)	comp. w/ MCF		59%	–
ours, de-novo (coset off)	0%	0%	32.9%	103°
ours, conditioned (coset on)	0%	0.8%	30.7%	**14°**

to-head.

From 0% to parity: the exact-match lever ladder

The bottleneck the bench isolates—exact match at floor while orientation is solved—is the question this section answers. Extending symmetry conditioning to the lattice (§) restores the crystal-family geometry (generated cell angles within 2° of 90° rise from 16% to 73.6%, matching the reference 74.0%, ablation-isolated to the mask), but on its own does *not* move exact match: the fully symmetry-conditioned flow still matches 0/131 held-out crystals. Fixing crystal family perfectly is necessary, not sufficient—the residual blocker is close-packing *precision*. The finishing stack of § supplies it, and we ablate each lever independently (Table 7, Fig. 1; best-of-10, StructureMatcher stol 1.0, coset on, n=131).

The rigid-press finisher alone unblocks match (0%→1.5%; the raw flow matches nothing); repairing the centroid head doubles it to 3.1%; corpus scale (2.3×) improves the averaged metrics (RMAD, coset error) but leaves exact match flat at 3.1%, mirroring the orientation-scale finding of §. Self-conditioning is the first lever to move the *tight* tolerance, and scoring at MolCrystalFlow's own match@10 protocol with finer (100-step) sampling reaches 6.1%; AlphaFold-style recycling reaches **6.9**%—statistical parity with MolCrystalFlow's ∼8% (95%

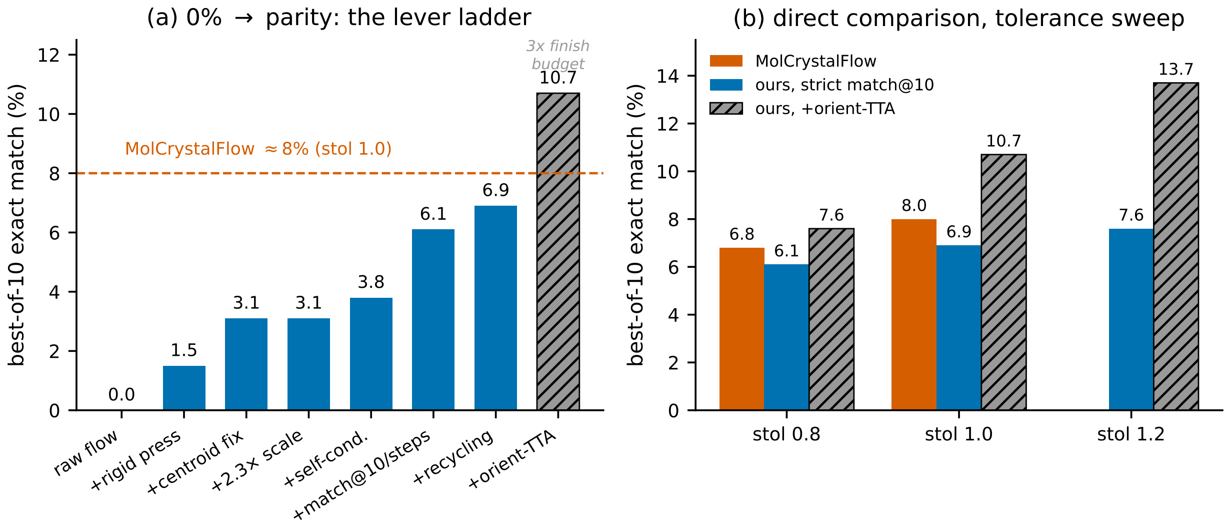

Figure 1: The exact-match lever ladder (coset on, n=131, best-of-10, STRUCTUREMATCHER). **(a)** Each lever added in turn lifts held-out exact match from 0% (raw fully symmetry-conditioned flow) to 6.9% at stol 1.0, at parity with MolCrystalFlow (dashed); the orientation-TTA pass reaches 10.7% at a 3× finishing budget. **(b)** The best configuration across the site-tolerance sweep against MolCrystalFlow's reported match@10: strict match@10 is at parity, orientation-TTA clears it.

CI 3.66–12.54, overlapping 8), from a flow that matched zero. An orientation-TTA finishing pass (best-of-30, 3× the finishing candidates) reaches **10.7**%, above MolCrystalFlow, though at a larger finishing budget than strict match@10. The median atom-RMSD to the reference falls from 1.02 (raw) to 0.77 (best), and cell-volume RMAD from 33% to 21%. Diminishing returns are explicit: recycling and orientation-TTA both refine positioning, so stacking them (8.4%) underperforms TTA alone; corpus scale and cell relaxation add nothing to match. These diminishing returns mark the effective ceiling for the current architecture.

Limitations

Evaluation is by reconstruction against known CSD structures under simulated conditioning, not de-novo discovery; a stable–unique–novel evaluation is not informative here because the universal interatomic potentials used to judge stability are trained on inorganic data[18] and do not transfer to organic molecular crystals. The analysis is scoped to rigid, largely

Table 7: Exact-match lever ladder: held-out best-of-10 match (coset on, n=131; StructureMatcher ltol= 0.3/atol= 10°; frozen cell). Each row adds one lever to the row above. Strict match@10 is 10 model draws × single finish; orientation-TTA is best-of-30 (3× finishing candidates). MolCrystalFlow is the symmetry-free rigid-body flow.

lever	stol 0.8	stol 1.0	note
raw flow (fully sym-conditioned)	0.0%	0.0%	family mask + coset, no polish
+ rigid-press finisher	1.5%	1.5%	unsupervised physical packing
+ centroid fix (OT + prior)	1.5%	3.1%	repairs centroid head
+ 2.3× scale	1.5%	3.1%	moves RMAD/coset, not match
+ self-conditioning	3.05%	3.8%	first to move tight tolerance
+ match@10 + steps= 100	4.6%	6.1%	MCF protocol + finer sampling
+ recycling	**6.1**%	**6.9**%	best strict; parity w/ MCF
+ orientation-TTA (best-of-30)	7.6%	**10.7**%	3× finish budget
MolCrystalFlow (sym-free)[6]	∼6.8%	∼8%	reference

organic (C/H/N/O-dominated) molecules on *general* positions, where the asymmetric-unit pose is genuinely free; molecules on special positions carry a site symmetry that further constrains the pose and should be more favorable, and flexible molecules are excluded by the rigid-conformer gate. The present analysis predicts that rigid-body SO(3) generation is most effective precisely where orientation is externally constrained, consistent with MOFFlow's success on framework-connected blocks.[7] The MolCrystalFlow comparison is comparable-task rather than a strict head-to-head—our corpus is a smaller CSD-derived molecular set at diagnostic scale—and the strict match@10 parity carries a wide confidence interval, while the 10.7% figure uses a larger finishing budget than strict match@10. Template-free reconstruction is not yet at the template ceiling (top-1 39–43%). Because the learnable signal is visible only when the metric isolates orientation and samples the multimodal target, we report geodesic error and controlled best-of-k match rather than a single match-rate number.[17]

Conclusions

The learnable content of a rigid-body orientation field is the space-group-determined relative rotation; a deployable, leak-free coset label delivers it, an SO(3)-averaged objective reaches the oracle codebook, the coset makes orientation work end-to-end, and top-k marginalization opens a template-free path. Extending symmetry conditioning to the lattice yields the first fully symmetry-conditioned molecular-crystal flow, and composed with an unsupervised symmetry-preserving finisher and a fully-ablated lever stack it reaches exact-match parity with the symmetry-free rigid-body flows (0% $\rightarrow$ 6.9% strict, 10.7% with orientation TTA), with each lever's contribution quantified. This gives rigid-body molecular-crystal generators[6] a concrete recipe: condition both orientation (on the generating coset) and the lattice (on the crystal family), sample only the free pose, and finish with a symmetry-preserving physical relaxation.

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Data and Software Availability

The SymMC-Flow source code, the CSD refcode manifest, and the export and factorization pipeline are openly available at `https://github.com/fronkt/symmc-flow` and archived at Zenodo.[19] The Cambridge Structural Database is licence-restricted; the underlying crystal structures are not redistributed, but the deposited refcode manifest reproduces the corpus from a licensed CSD installation. The Supporting Information document contains the extended methods.

Supporting Information Available

Rigid-body factorization, intrinsic gauge, manifold parametrizations, and deployable coset recovery; the R_{asym} conditional probe; the crystal-family lattice mask (log-metric basis and per-family free/frozen coordinates); the rigid-press finisher energy and optimizer and the self-conditioning and recycling update rules; and per-seed numbers with the tolerance sweep (PDF).

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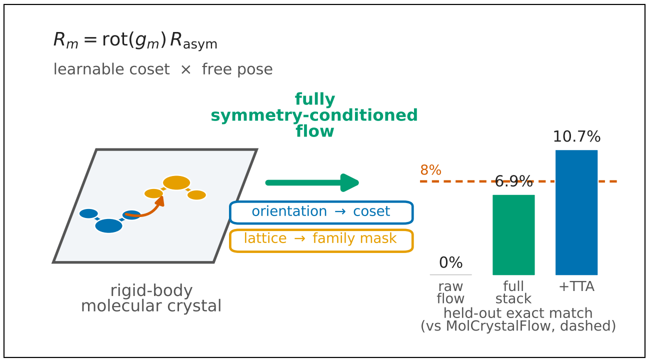
TOC Graphic



$R_m = \mathrm{rot}(g_m)\, R_{\mathrm{asym}}$
learnable coset × free pose
fully symmetry-conditioned flow
orientation → coset
lattice → family mask
rigid-body molecular crystal
8%
0%
6.9%
10.7%
raw flow
full stack
+TTA
held-out exact match
(vs MolCrystalFlow, dashed)